MOSTAFA HUSSEIN

Robot Learning · VLA & Generative Policies · Continual Learning · Perception
mostafa.hussein@unh.edu | https://mostafa-hussein.github.io/ | linkedin.com/in/mostafa-hussein-phd

SUMMARY

Senior robot learning researcher (PhD in Robotics, University of New Hampshire) with 10+ years of experience spanning robot learning, manipulation, perception, autonomous systems, and real-world deployment. Specializes in learning visuomotor policies for long-horizon tasks, combining imitation learning, behavior cloning, Dagger, residual policy learning, flow matching, reinforcement learning, and Vision-Language-Action (VLA) models. Experience spans research prototypes and production robotics, including real-robot policy evaluation, multimodal data, large-scale training, and deployment. Strong record of turning research ideas into deployable systems with measurable gains in accuracy, robustness, sample efficiency, and task reliability. Previously shipped production perception models at Amazon Robotics across U.S. fulfillment centers and led multi-month in-home robot deployments in NIH-funded trials. At Persona AI I lead a research program on continual policy improvement for VLA foundation models like (NVIDIA GR00T, π0.5/π0.7) on long-horizon bimanual manipulation — how a deployed policy keeps improving overtime with minimal training. First author at ICRA, IROS, and CoRL.

SKILLS

Robot Learning	VLA models (GR00T, π0-π0.7, OpenVLA); generative policies (flow matching, Diffusion Policy, DP3); imitation learning (BC, robust LfD, DAgger); RL (residual, sample efficient, online/offline); continual learning; policy evaluation and motion-quality metrics.
ML Infrastructure	PyTorch; multi-GPU distributed training (torchrun); training multi-billion-parameter models; Weights & Biases; LeRobot datasets; MCAP / ROS-bag data pipelines; dataset curation; Docker, Linux, Git.
Perception	VIT, CNNs, PointNet, 3D point clouds, depth sensing, self-supervised & contrastive learning (DINO)
Robotics & Code	Python, C++; ROS 2, Nav2, Behavior Trees; teleoperation data collection; multi-camera fusion; Jetson edge deployment; Franka FR3, humanoids, Stretch 3, YuMi, NAO; MuJoCo; Isaac Sim / Isaac Lab.

EXPERIENCE

Senior Deep Learning Manipulation Engineer | Persona AI, Houston, TX — May 2026 – Present

- Develop deep learning manipulation policies for **long-horizon, multi-stage tasks**, with emphasis on robust execution, recovery, task generalization, and reducing policy-update cost with the goal of deploying it on humanoid robots in **industrial settings (factories, shipyards)**.
- Prototype complementary policy-learning strategies including behavior cloning, DAgger, reinforcement learning, residual policy learning, sample-efficient RL, and vision-language-action models such as GR00T.
- Design **continual-learning workflows** that incorporate new demonstrations, failure cases, and deployment rollouts into policy updates while preserving previously learned skills using techniques such **Experience Replay (ER).**
- Build evaluation protocols around success rate, episode/sample efficiency, trajectory length, motion smoothness (jerk), and cycle time.

Applied Scientist II | University of New Hampshire (CARL), Durham, NH — Jan 2025 – April 2026

- Co-advised the research behind **Trajectory-Consistent Flow Matching** (IROS 2026), which closes the train–inference gap in flow-matching policies with multi-step trajectory-consistency training, velocity-smoothness regularization, and RK4 inference. On real robots the policy reaches **70% and 60% success on two long-horizon multi-phase tasks where Diffusion Policy and Consistency-FM baselines score 0%**, and 100% on precision tool placement on a Boston Dynamics Spot.
- Led development and deployment of autonomous assistive robots in **four private homes for 1–6 months** each (NIH-funded trials): designed the **ROS 2 / Nav2 / Behavior-Tree** autonomy stack (**CaRE-BT**, ICSR 2025) fusing multi-camera tracking, activity and face recognition; **500+ scheduled care interventions** over six months of field operation, co-authored the co-design study of the in-home trials (IJERPH 2026).

Applied Scientist II | Amazon Robotics, Boston, MA Sep 2022 – Dec 2024

- Delivered the **production perception foundational model for the Sparrow bin-picking robot**: self-supervised **ViT** embeddings (**DINO**) with **CLIP** image–text alignment and domain-invariant representations — to handle incomplete reference databases, occlusion, deformation, and packaging variability; raising identification accuracy from **80% to 98.5%** and cutting Damage-Per-Million from **2000 to 1000**; deployed to stations across U.S. fulfillment centers (ICRA 2024).

Research Assistant | Cognitive Assistive Robotics Lab (CARL), UNH 2017 – 2022

- Developed **Robust Maximum-Entropy Behavior Cloning** with automatic **adversarial-demonstration detection** for safe imitation learning on multi-step tasks (CoRL 2022, IROS 2021, NeurIPS-W 2020), validated on a YuMi tea-making task; designed **IRL in POMDPs** for learning interaction policies from demonstrations (ICRA 2019), applied to robot-mediated interventions for children with autism.
- Internships: **Amazon Robotics AI** (2021, deep learning for perception/manipulation); **MERL** (2021, contact models in physics engines for robotic transfer learning); **UNH RL & Robustness Lab** (2020, robust imitation learning).

Research & Teaching Assistant | Assiut University, Egypt 2011 – 2017

- Built a ROS learning-from-demonstration stack (**GMM motion primitives with Variational Bayes**, motif discovery) that taught a NAO robot to write in English and Arabic (ICMA 2017, IEA/AIE 2015).

EDUCATION

- **Ph.D., Computer Science (Robotics)** — University of New Hampshire, Durham, NH 2017 – 2023
  Thesis: *Robust Behavior Cloning for Multi-Step Sequential Task Learning by Robots* (advisor: Prof. Momotaz Begum).
- **M.Sc., Electrical Engineering (Computer & Control)** — Assiut University, Egypt (top of class) 2011 – 2016
- **B.Sc., Electrical Engineering (Computer & Control)** — Assiut University, Egypt (top of class) 2005 – 2010

SELECTED PROJECTS

Continual Policy Learning for Long-Horizon Bimanual Manipulation | Persona AI 2026 – Present

Tech: VLA training (GR00T N1.7, π0/π0.5) · Imitation learning · DAgger · Residual RL · Experience replay · Value monitors · LeRobot

- Reference task: a four-zone tray-assembly line — the right arm picks, places, and stacks trays while the left arm inserts cylinders, over repeated cycles. Base policy: a fully fine-tuned GR00T N1.7 VLA trained on teleoperated demonstrations, with count-agnostic, arm-assignment-free prompts so the language conditioning holds across cycles. The main problem is that long-horizon task compounds small policy errors across many sub-tasks, and policy retraining can become expensive when failures appear after deployment.
- Improvement loop in escalating levels: (A) full fine-tuning on all episodes per cycle only for reference; (B) targeted data on failed sub-tasks with a controlled comparison of sequential fine-tuning, experience replay, and retraining from base; (C) DAgger on-policy correction; (D) residual RL on the frozen VLA. While the loop is progressing, we use the head-camera to train a monitor model (binary → per-subtask → continuous value) supplies success/value signals for RL and RECAP-style (π0.6) training.
- Defined the metrics taxonomy (outcome, progress, efficiency, motion quality) and the frozen 20-rollout panel used to compare every checkpoint; built the LeRobot pipeline, the offline evaluation harness, and smoothness metrics (jerk, SPARC).

Trajectory-Consistent Flow Matching for Robust Visuomotor Policy Learning | UNH • IROS 2026 2025 – 2026

Tech: Flow matching · Consistency-FM · Diffusion Policy · Dual-view PointNet encoder · RK4 inference · Franka FR3 & Boston Dynamics Spot

- Research project addressing error accumulation in flow-matching policies for long-horizon manipulation, where pointwise velocity training can diverge from the integrated trajectory executed at inference time.
- Contributed to a trajectory-level learning framework combining multi-step trajectory consistency, velocity-field regularization, auxiliary rectified-flow supervision, and higher-order RK4 inference.
- Evaluated the approach on real-robot manipulation tasks across Franka and Boston Dynamics Spot and on MetaWorld simulation tasks; the reported method reached 70% and 60% success on two long-horizon multi-phase tasks where standard baselines achieved 0%, and 100% on precision tool placement.

Scaling Object-Centric Robotic Manipulation with Multimodal Object Identification | Amazon Robotics • ICRA 24 2022 – 2024

Tech: DINO, CLIP, Vision Transformers, Contrastive Learning, Multimodal Retrieval, 190K+ SKU References

- Warehouse manipulation must identify an open-ended object set despite incomplete reference databases, occlusion, deformation for that I developed a multimodal object-identification foundational model, combining image and text representations to generalize across incomplete references, occlusion, deformation, and packaging variation.
- Deployed the resulting perception system across Sparrow stations in U.S. fulfillment centers, improving identification accuracy from 80.5% to 98.5%.

CaRE-BT — Assistive Robots in Real Homes | UNH · ICSR 2025, IJERPH 2026 2025 – 2026

Tech: ROS 2 · Nav2 · Behavior Trees · PDDL planning · multi-camera tracking · Face recognition / re-ID · Activity recognition · IoT · Stretch 3 · Edge deployment

- Care robots must make reliable sequential decisions from noisy perception while human routines and home conditions change
  over time. As team lead and principal researcher, built the open-source framework that turns symbolic care protocols (medication, exercise, safety checks) into executable behavior trees so a mobile robot adapts to people and changing homes in real time: a situational-awareness layer fusing camera tracking, face re-ID and IoT signals, and a bi-level planner that replans over a self-healing knowledge base and dispatches navigation, docking, speech and call services on embedded hardware.
- The system was deployed on Stretch 3.0 and related platforms in real homes; field studies included multi-month deployments, with 500+ scheduled reminders logged over six months in the broader smart-home program.

Robust Behavior Cloning for Multi-Step Sequential Task Learning | UNH • CORL 22, IROS 21 2018-2022

Tech: Behavior Cloning, Maximum Entropy RL, Robust LfD, IRL, Adversarial Demonstration Detection

- Developed Robust Maximum-Entropy Behavior Cloning to identify and down-weight incorrect or unsafe demonstrations, enabling policies to learn from imperfect human data.
- Validated robust policy learning in simulation and on real robotic tasks, including a YuMi tea-making task, improving resilience and sample efficiency.

Learning Interaction Dynamics and Policies from Human Demonstrations | UNH · ICRA 2019 2018 – 2019

- In human–robot interaction the reward is never given, and the state is only partially observable. Recovered the reward functions underlying sequential interaction tasks by reducing the POMDPs to tractable MDPs and applying **IRL**; validated on robot-mediated social-greeting and object-naming interventions with children with autism.

Autonomous Learning of Perceptual and Motion Primitives | Assiut University · ICMA 2017, IEA/AIE 2015 2013 – 2017

- A ROS stack that discovers the primitives itself: **motif discovery** finds repeated actions within demonstrations, and **GMM motion primitives with Variational Bayes** select the number of components automatically. Taught a NAO robot to write in English and Arabic — two scripts with very different stroke structure, from one pipeline.

SELECTED PUBLICATIONS

[1] R. Ahmed, S. Nag, M. Akash, **M. Hussein**, M. Begum. Trajectory-Consistent Flow Matching for Robust Visuomotor Policy Learning. **IROS 2026**.

[2] O. Ghattas, **M. Hussein**, M. Alam, M. Akash, S. Arthanat, D. LaRoche, M. Begum. CaRE-BT: An Embodied Planning Framework for In-Home Assistive Robots. **ICSR 2025**.

[3] C. Mitash, **M. Hussein**, J. Vanbaar, V. Terhuja, K. Katyal. Scaling Object-centric Robotic Manipulation with Multimodal Object Identification. **ICRA 2024**.

[4] **M. Hussein**, M. Begum. Detecting Incorrect Visual Demonstrations for Improved Policy Learning. **CoRL 2022**.

[5] **M. Hussein**, B. Crowe, M. Clark-Turner, M. Petrik, M. Begum. Robust Behavior Cloning with Adversarial Demonstration Detection. **IROS 2021**.

[6] **M. Hussein**, B. Crowe, M. Clark-Turner, M. Petrik, M. Begum. Robust Maximum Entropy Behavior Cloning. Workshop on Robot Learning: Control and Interaction in the Real World, **NeurIPS 2020.**

[7] **M. Hussein**, M. Begum, M. Petrik. Inverse Reinforcement Learning of Interaction Dynamics from Demonstrations. **ICRA 2019**.

[8] **M. Hussein**, Y. Mohammad, S. A. Ali. COLD: A ROS Package for Continuous Learning from Demonstration Teaching a Robot to Write. **ICMA 2017.**

[9] **M. Hussein**, Y. Mohammad, S. A. Ali. Learning from Demonstration Using Variational Bayesian Inference. **IEA/AIE 2015.**